

Proving: "The trace of an idempotent matrix equals the rank of the matrix"

How could we prove that the "The trace of an idempotent matrix equals the rank of the matrix"?

This is another property that is used in my module without any proof, could anybody tell me how to prove this one?

(linear-algebra) (matrices)

asked Jan 23 '12 at 2:48



Quixotic

11.2k 15 74 159

3 What is your *module*? – Mariano Suárez-Álvarez ♦ Jan 23 '12 at 2:53

4 Answers

An idempotent has two possible eigenvalues, zero and one, and the multiplicity of one as an eigenvalue is precisely the rank. Therefore the trace, being the sum of the eigenvalues, is the rank (assuming your field contains $\mathbb{Q}$...)

answered Jan 23 '12 at 2:54



Mariano Suárez-Álvarez ♦

107k 7 145 263

5 Just in case it isn't clear, the reason the eigenvalues are 0 and 1 is because all the eigenvalues are roots of the minimal polynomial, which is $x^2 - x$. Because the minimal polynomial has no repeated roots, it is diagonalizable, and thus has a basis of eigenvectors. Writing the vector space as $V_0 \oplus V_1$, the transformation is projection onto V_1 , and so the rank is the dimension of V_1 . – Aaron Jan 23 '12 at 3:49

4 Just for the record, you don't need minimal polynomials or eigenvectors. Let $A : V \rightarrow V$ be idempotent, let $V_0 = \text{Ker}(A)$ and $V_1 = \text{Im}(A)$. If $u \in V_0 \cap V_1$ then $u = Au = 0$, so $V_0 \cap V_1 = \{0\}$. For any v , we have $v = Av + (v - Av)$, and $Av \in V_1$, $v - Av \in V_0$, so $V = V_0 + V_1$. So we have $V = V_0 \oplus V_1$. (Probably not the right route for most students, but I happen to be teaching a class at the moment where I want this fact and we haven't hit Jordan canonical form yet.) – David E Speyer Sep 12 '14 at 13:03

I came to this page by accident but just wanted to note that the statement above that

"the multiplicity of one as an eigenvalue is precisely the rank"

is non-trivial and is not true for general matrices. You still need to prove that algebraic multiplicity equals geometric multiplicity (in other words, that the number of linearly independent eigenvectors equals the multiplicity of one)

The fact that "since $y = Px = P(Px)$ therefore members of an orthogonal basis of the range of P are also eigenvectors of P " is the missing piece. Because of this we can comfortably say that the rank is at least equal to the multiplicity. After that we need to state that none of the eigenvectors whose eigenvalue is zero could contribute to the range (though that one might omit because it's trivial.) @DavidSpeyer said similar things in his comment.

answered Oct 22 '14 at 2:34



Pushpendre

287 2 13

Thanks for the clarification. That statement caused confusion for a while. – Prince Kumar Dec 3 '16 at 12:36

1 In the same sentence as that statement is made I clearly stated that I was taking about idempotent matrices. Confusion could only arise if you read only *half* of what I wrote. – Mariano Suárez-Álvarez ♦ Feb 25 '17 at 17:54

1 @MarianoSuaÁlvarez A wumpus has two clubs and a club's multiplicity is precisely the rank. Is it

really all that clear that this statement about a club's multiplicity only holds when the club is wielded by a wumpus? – [Pushpendre](#) Feb 25 '17 at 22:53

To be honest, I have no idea what that has to do with the fact that the claim I made was made in a context which I was careful to make explicit. You are of course free to... hmm... I don't know what you are doing... but sincerely I could not care less. – [Mariano Suárez-Álvarez](#) ♦ Feb 25 '17 at 22:56

Very well, fine by me. – [Pushpendre](#) Feb 25 '17 at 23:06

Sorry to post solution to this such a old question, but "The trace of an idempotent matrix equals the rank of the matrix" is very basic problem and every answer here is using the solution using **eigen values**. But there is another way which should be highlighted.

Solution:

Let $A_{n \times n}$ is a idempotent matrix. Using [Rank factorization](#), we can write $A = B_{n \times r} C_{r \times n}$ where B is of full column rank and C is of full row rank, then B has left inverse and C has right inverse.

Now, since $A^2 = A$, we have $BCBC = BC$. Note that,

$$BCBC = BC \Rightarrow CBC = C \Rightarrow CB = I_{r \times r}$$

Therefore

$$\text{trace}(A) = \text{trace}(BC) = \text{trace}(CB) = \text{trace}(I_{r \times r}) = r = \text{rank}(A) \quad \blacksquare$$

edited Jul 4 '17 at 9:33

answered Jul 4 '17 at 6:26



MAN-MADE

3,356 1 7 28

+1 Thanks for posting this, it is a very pretty argument. – [Erick Wong](#) Jul 4 '17 at 7:10

@MANMAID Please explain a little more why $CBC = C \Rightarrow CB = I_{r \times r}$? C is not invertible.. – [Widawensen](#) Jul 4 '17 at 9:17

1 @Widawensen C is not invertable, but C has right inverse. – [MAN-MADE](#) Jul 4 '17 at 9:35

@MANMAID And I suppose B has left inverse...Is it some kind of theorem which says us about this right (left) inverse ? I would be grateful for extending my knowledge.. – [Widawensen](#) Jul 4 '17 at 9:40

@Widawensen In the book D.A. **HARVILLE-Matrix Algebra from a Statistician's Perspective**, chapter 8, you can find it elaborately. This chapter is only about inverses actually. – [MAN-MADE](#) Jul 4 '17 at 9:53

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Hint: what are the eigenvalues of an idempotent matrix?

answered Jan 23 '12 at 2:53



Gerry Myerson

139k 7 135 283